

Notes: Mayzlin, Dover & Chevalier (AER, 2014)
Promotional Reviews: An Empirical Investigation of Online Review Manipulation

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1 Big picture

- **Question.** Can we detect promotional or fake online reviews empirically without classifying individual reviews as fake, and what firm/market conditions create incentives to manipulate reviews?
- **Setting.** The paper compares hotel reviews on two review platforms with different identity-verification rules: TripAdvisor, where anyone can post, and Expedia, where only customers who booked a stay through Expedia can post.
- **Core identification idea.** If fake reviewing is cheaper on TripAdvisor than on Expedia, then the difference in a hotel's review distribution across the two sites should vary systematically with the hotel's incentives to manipulate reviews.
- **Main dependent variables.** The paper studies the TripAdvisor-minus-Expedia difference in the share of negative reviews (one- and two-star) and the share of positive reviews (five-star).
- **Main incentive variables.** Review manipulation incentives vary with whether the hotel is independent versus chain-affiliated, whether the owner is a single-unit or multiunit owner, whether the hotel has nearby competitors, and whether those competitors are independent or multiunit-owned.
- **Main results.** Hotels with high incentives to promote themselves have more five-star TripAdvisor reviews relative to Expedia; neighboring competitors of hotels with high incentives to attack have more one- and two-star TripAdvisor reviews relative to Expedia.
- **Organizational insight.** Large chains, multiunit owners, and multiunit management companies appear less likely to manipulate reviews, consistent with higher reputational cost and better internal controls.
- **Economic takeaway.** Review manipulation is not random noise. It varies predictably with market structure, platform design, and organizational form.

2 Incremental contribution of the paper

- **Moves fake-review detection beyond text classification.** Prior computer-science work often tried to detect fake reviews from linguistic markers. Mayzlin, Dover, and Chevalier instead use cross-platform institutional differences and economic incentives.
- **Uses platform design as a quasi-verification contrast.** The paper exploits a sharp difference in posting cost: TripAdvisor allows anyone to review, while Expedia requires a verified booking. This turns platform design into an empirical source of variation.
- **Introduces an incentive-based detection strategy.** The paper does not need to know whether any particular review is fake. It looks for systematic differences in review distributions where incentives to fake should be high.
- **Connects review manipulation to organizational form.** Independent hotels and single-unit owners are predicted to face lower reputational costs of manipulation than branded chains and multiunit owners. This links digital fraud to the economics of franchising and organizational control.
- **Separates positive and negative manipulation.** The paper studies both fake positive reviews for one's own hotel and fake negative reviews aimed at neighboring competitors.
- **Uses local competition to identify negative manipulation.** Hotels with close neighbors have stronger incentives to attack rival hotels. The identity of the neighbor matters, which strengthens the economic mechanism.
- **Provides policy implications for review platforms.** Verification rules reduce manipulation but also reduce the number of reviews. The paper therefore highlights a tradeoff between review quantity and review integrity.
- **Broad contribution.** The paper creates a template for detecting hidden misconduct by combining platform rules, economic incentives, and difference-in-differences logic.

3 Data and measurement

3.1 Platforms and review data

- Reviews are scraped in October 2011 from TripAdvisor and Expedia.
- The sample includes hotels in the 25th to 75th largest US cities by population, excluding especially dense or unusual travel markets.
- The authors match hotels across TripAdvisor, Expedia, and Smith Travel Research (STR) using names and addresses.
- The final matched sample contains 2,931 hotels that have reviews on both TripAdvisor and Expedia.
- TripAdvisor has many more reviews per hotel than Expedia because TripAdvisor does not require verified purchase.

Interpretation: TripAdvisor and Expedia provide the same type of object–hotel reviews—but differ in the cost of faking. This contrast is the backbone of the design.

3.2 Organizational and market structure data

- STR provides hotel brand affiliation, ownership, management, class/tier, age, and location characteristics.
- A hotel is classified as independent if it has no chain affiliation.
- Ownership distinguishes single-unit owners from multiunit owners.
- Management distinguishes small management companies from multiunit management companies.
- Neighbor variables use hotels within a 0.5 km radius in the main specifications.

Interpretation: The organizational variables proxy for the cost of manipulation. Larger organizations are assumed to bear higher reputational costs and to have better control systems.

3.3 Main review measures

For hotel i in city j , the main dependent variables compare the share of an N -star review category across TripAdvisor and Expedia:

$$Y_{ij}^N = \frac{NStarReviews_{ij}^{TA}}{TotalReviews_{ij}^{TA}} - \frac{NStarReviews_{ij}^{Exp}}{TotalReviews_{ij}^{Exp}}.$$

- Negative manipulation is measured mainly with the TripAdvisor-minus-Expedia difference in one- and two-star review share.
- Positive manipulation is measured mainly with the TripAdvisor-minus-Expedia difference in five-star review share.
- One-time contributors are studied as a suspicious reviewer type, because creating a new TripAdvisor identity is cheap.

Interpretation: A simple comparison of TripAdvisor and Expedia reviews would be contaminated by different user populations. The paper therefore studies how the TripAdvisor-Expedia gap varies across hotel incentives.

4 Models and empirical specifications

4.1 Baseline difference-in-differences style regression

The main estimating equation is:

$$Y_{ij}^N = X_{ij}\beta_1 + OwnAf_{ij}\beta_2 + Nei_{ij}\beta_3 + NeiOwnAf_{ij}\beta_4 + \sum_j \gamma_j + \varepsilon_{ij}.$$

- X_{ij} contains hotel controls such as site rating, hotel age, all suites, convention center, restaurant, tier controls, and location controls.
- $OwnAf_{ij}$ contains own-hotel affiliation and ownership variables.
- Nei_{ij} indicates whether the hotel has nearby competitors.
- $NeiOwnAf_{ij}$ measures the ownership and affiliation of the hotel neighbors.
- City fixed effects absorb city-level differences in review behavior and hotel markets.

Interpretation: The coefficient on own independence and ownership captures positive self-promotion incentives. The coefficient on neighbor independence or single-unit ownership captures negative manipulation by rival hotels.

4.2 Theoretical manipulation model

In the appendix, two horizontally differentiated firms choose positive self-promotion and negative rival-targeting review effort. A stylized result is:

$$e_{A,A}^* = \frac{p_A}{2\delta_A t}, \quad e_{A,B}^* = \frac{p_A}{2\gamma_A t}, \quad e_{B,B}^* = \frac{p_B}{2\delta_B t}, \quad e_{B,A}^* = \frac{p_B}{2\gamma_B t}.$$

- $e_{A,A}^*$ is firm A's fake positive effort for itself.
- $e_{A,B}^*$ is firm A's fake negative effort against firm B.
- δ and γ are reputational/manipulation cost parameters.
- Higher manipulation costs reduce manipulation effort.

Interpretation: The appendix model gives the core comparative static: lower reputational cost leads to more manipulation, including both positive self-promotion and negative attacks on competitors.

4.3 One-time reviewer specification

The one-time reviewer table replaces the dependent variable with measures based on reviews written by TripAdvisor users who only posted one review. This tests whether the patterns are stronger for a suspicious review type.

Interpretation: One-time reviewers are not proof of fraud, but they are a plausible proxy for low-cost manipulation.

4.4 Robustness specifications

Robustness checks include:

- dropping controls to assess selection on observables;
- adding chain fixed effects;
- varying neighbor radius from 0.3 km to 0.9 km;
- focusing on one-star reviews versus one- and two-star reviews;
- comparing TripAdvisor with Orbitz in the online appendix.

Interpretation: The strongest evidence is for negative manipulation against competitors. Positive manipulation appears, but is more sensitive to selection concerns.

5 Identification and empirical design

5.1 Core identification problem

Promotional reviews are hidden by design. A review can be extreme because it is fake or because a real customer had an extreme experience. The paper avoids classifying individual reviews and instead studies systematic review-distribution differences across platforms and incentive conditions.

5.2 Why two platform comparison helps

TripAdvisor and Expedia differ in verification cost. Anyone can post on TripAdvisor, but Expedia requires a recent booking. If hotels fake reviews, manipulation should be cheaper on TripAdvisor than on Expedia.

Interpretation: This allows the authors to use Expedia as a harder-to-manipulate benchmark for each same hotel.

5.3 Why cross-sectional incentives matter

A difference between TripAdvisor and Expedia could reflect different reviewer populations. The paper therefore asks whether that difference is larger exactly where economic incentives to manipulate are larger.

- A hotel with nearby rivals has stronger incentives to post fake negative reviews.
- Independent or single-unit-owned hotels have lower reputational cost and weaker organizational controls.
- Branded chains and multiunit owners/managers face higher reputational spillovers.

5.4 Main identifying assumption

The key assumption is that TripAdvisor and Expedia users do not differentially value hotel ownership, affiliation, and neighbor characteristics in a way that mimics the manipulation predictions. The authors add rich hotel controls and robustness tests to reduce this concern.

5.5 Limits of the design

- Manipulation is not directly observed.
- The design cannot identify which individual reviews are fake.
- Expedia users and TripAdvisor users may differ in unobservable preferences.
- The paper cannot measure the sales effect of manipulation.

- Textual content analysis is not part of the main identification strategy.

Interpretation: The paper’s credibility comes from a coherent set of predictions across own hotel traits, neighbor traits, one-time reviewers, radius robustness, and site-verification contrasts.

6 Section-by-section study guide

- **Introduction.** Explains why fake reviews damage consumer learning and why direct fake-review classification is hard.
- **Prior literature.** Positions the paper relative to advertising, persuasion, promotional chat, fake-review detection, and organizational-form literatures.
- **Data.** Describes TripAdvisor, Expedia, STR hotel characteristics, ownership/affiliation, location, and neighbor construction.
- **Theory.** Provides the cost-of-manipulation logic: lower reputational costs imply more positive and negative fake reviewing.
- **Empirical results.** Tests review-distribution differences across own hotel and neighbor incentive variables.
- **Robustness.** Uses one-time reviewers, no-controls specifications, negative-review variants, chain fixed effects, and alternative platforms/radii.
- **Conclusion.** Discusses implications for review-system design, organization, fake-review detection, and future work.

7 Tables

7.1 Tables 1 and 2: Review characteristics and hotel organization

Read:

- Table 1 shows TripAdvisor has many more reviews per hotel than Expedia: 119.58 versus 42.16 on average.
- TripAdvisor ratings are lower on average than Expedia ratings: 3.52 versus 3.95.
- TripAdvisor has higher shares of one-star and two-star reviews than Expedia, while Expedia has more five-star reviews.
- Table 2 shows 17 percent of hotels are independent, 31 percent have multiunit owners, and 52 percent have multiunit management companies.
- 76 percent of hotels have a neighbor within 0.5 km; 31 percent have an independent neighbor and 49 percent have a multiunit owner neighbor.

Detailed interpretation: Table 1 defines the platform gap. TripAdvisor has more reviews and a more negative distribution. Table 2 defines the incentive variables. The paper later asks whether these organizational and neighbor variables predict the TripAdvisor-minus-Expedia review gap.

Economic message: Review manipulation is studied as the interaction of platform design and economic incentives. Tables 1 and 2 define both sides of that interaction.

TABLE 1—USER REVIEWS AT TRIPADVISOR AND EXPEDIA

	Mean	Standard deviation	Minimum	Maximum
Number of TripAdvisor reviews	119.58	172.37	1	1,675
Number of Expedia reviews	42.16	63.24	1	906
Average TripAdvisor star rating	3.52	0.75	1	5
Average Expedia star rating	3.95	0.74	1	5
Share of TripAdvisor one-star reviews	0.14			
Share of TripAdvisor two-star reviews	0.11			
Share of Expedia one-star reviews	0.07			
Share of Expedia two-star reviews	0.08			
Share of TripAdvisor five-star reviews	0.31			
Share of Expedia five-star reviews	0.44			
Total number of hotels	2,931			

Note: The table reports summary statistics for user reviews for 2,931 hotels with reviews at both TripAdvisor and Expedia collected in October of 2011.

(a) Table 1: User Reviews at TripAdvisor and Expedia

TABLE 2—OWN AND NEIGHBOR HOTEL AFFILIATION, OWNERSHIP, AND MANAGEMENT AND STRUCTURE

Hotel status	Share of all hotels with reviews	Share of independent hotels	Share of chain affiliated hotels
Independent	0.17	1.00	0.00
Marriott Corporation affiliate	0.14	0.00	0.17
Hilton Worldwide affiliate	0.12	0.00	0.15
Choice Hotels International affiliate	0.11	0.00	0.13
Intercontinental Hotels Group affiliate	0.08	0.00	0.10
Best Western Company affiliate	0.04	0.00	0.04
Multiunit owner	0.31	0.16	0.34
Multiunit management company	0.52	0.35	0.55
Multiunit owner AND multiunit management company	0.26	0.12	0.29
Hotel has a neighbor	0.76	0.72	0.77
Hotel has an independent neighbor	0.31	0.50	0.27
Hotel has a multiunit owner neighbor	0.49	0.52	0.49
Hotel has a multiunit management entity neighbor	0.59	0.58	0.59
Total hotels in sample = 2,931			

Note: Table shows summary information about brand affiliation, ownership, and management characteristics for 2,931 hotels associated with reviews at TripAdvisor and Expedia collected in October of 2011.

(b) Table 2: Hotel Affiliation, Ownership, and Management Structure

7.2 Tables 3 and 4: Baseline manipulation estimates and management-company specification

Read:

- Table 3 estimates the baseline equation for one-/two-star and five-star review gaps.
- Having any neighbor raises the one-/two-star TripAdvisor-minus-Expedia gap.
- Having an independent neighbor raises negative reviews; having a multiunit-owner neighbor lowers them.
- Independent own hotels have more five-star TripAdvisor reviews; multiunit-owned hotels have fewer five-star TripAdvisor reviews.
- Table 4 adds multiunit management company indicators and shows that management companies also reduce apparent positive manipulation.

Detailed interpretation: Tables 3 and 4 are the main evidence. The negative-review results are especially strong: nearby hotels with low-cost manipulation incentives generate more negative TripAdvisor reviews for rivals. The five-star results support positive self-promotion but are somewhat more sensitive to selection concerns.

Economic message: Hotels appear to manipulate both sides of the review distribution: attacking competitors with negative reviews and boosting themselves with positive reviews.

TABLE 3—ESTIMATION RESULTS OF EQUATION (1)

		Difference in share of one- and two-star reviews	Difference in share of one- and two-star reviews	Difference in share of five-star reviews
X_{ij}	Site rating	-0.0067 (0.0099)	-0.0052 (0.0099)	-0.0205** (0.0089)
	Hotel age	0.0004*** (0.0002)	0.0003* (0.0002)	0.0002 (0.0002)
	All suites	0.0146 (0.0092)	0.0162* (0.0092)	0.0111 (0.0111)
	Convention center	0.0125 (0.0086)	0.0159* (0.0091)	-0.0385*** (0.0113)
	Restaurant	0.0126 (0.0093)	0.0114 (0.0092)	0.0318*** (0.0099)
	Hotel tier controls?	Yes	Yes	Yes
	Hotel location controls?	Yes	Yes	Yes
$OwnAf_{ij}$	Hotel is independent		0.0139 (0.0110)	0.0240** (0.0103)
	Multiunit owner		-0.0011 (0.0063)	-0.0312*** (0.0083)
Nei_{ij}	Has a neighbor	0.0192** (0.0096)	0.0296** (0.0118)	-0.0124 (0.0119)
$NeiOwnAf_{ij}$	Has independent neighbor		0.0173* (0.0094)	-0.0051 (0.0100)
	Has multiunit owner neighbor		-0.0252*** (0.0087)	-0.0040 (0.0097)
γ_j	City-level fixed effects?	Yes	Yes	Yes
	Observations	2,931	2,931	2,931
	R^2	0.05	0.06	0.12

Notes: Regression estimates of equation (1). The dependent variable in all specifications is the share of reviews that are N star for a given hotel at TripAdvisor minus the share of reviews for that hotel that are N star at Expedia.

(a) Table 3: Estimation Results of Equation (1)

TABLE 4—MANAGEMENT COMPANY SPECIFICATIONS

		Difference in share of one- and two-star reviews	Difference in share of five-star reviews
X_{ij}	Site rating	-0.0047 (0.0100)	-0.0183** (0.0090)
	Hotel age	0.0003** (0.0002)	0.0002 (0.0002)
	All suites	0.0169* (0.0091)	0.0144 (0.0112)
	Convention center	0.0163* (0.0090)	-0.0363*** (0.0113)
	Restaurant	0.0110 (0.0092)	0.0323*** (0.0099)
	Hotel tier controls?	Yes	Yes
	Hotel location controls?	Yes	Yes
$OwnAf_{ij}$	Hotel is independent	0.0141 (0.0111)	0.213** (0.0104)
	Multiunit owner	-0.0014 (0.0065)	-0.0252*** (.0086)
	Multiunit management company	0.0022 (0.0077)	-0.0211** (0.0091)
Nei_{ij}	Has a neighbor	0.0379*** (0.0142)	-0.0098 (0.0140)
$NeiOwnAf_{ij}$	Has independent neighbor	0.0173* (0.0094)	-0.006 (0.0100)
	Has multiunit owner neighbor	-0.0169* (0.0097)	0.0004 (0.0114)
	Has multiunit management company neighbor	-0.0183 (0.0125)	-0.0059 (0.0136)
γ_j	City-level fixed effects?	Yes	Yes
	Observations	2,931	2,931
	R^2	0.05	0.06

(b) Table 4: Management Company Specifications

7.3 Tables 5 and 6: One-time contributors and no-controls robustness

Read:

- Table 5 studies reviews written by one-time TripAdvisor contributors.
- Independent hotels have a substantially higher share of one-time contributor reviews.
- Multiunit owners have fewer one-time contributor reviews and fewer five-star reviews.
- Table 6 re-estimates the core specifications without controls.
- The no-controls results are broadly similar to baseline results, especially for neighbor variables.

Detailed interpretation: Table 5 uses a suspicious-review proxy. It is not conclusive, but its patterns align with the main theory. Table 6 addresses omitted-variable concern using the Altonji-Elder-Taber logic: if adding controls barely attenuates coefficients, unobservables would need to be very strong to explain the results.

Economic message: The main results survive a suspicious-review proxy and a no-controls comparison, strengthening the interpretation of review manipulation.

TABLE 5—RESULTS FOR TRIPADVISOR ONE-TIME CONTRIBUTOR REVIEWERS

		Share of one-time contributor user reviews	Difference in share of one- and two-star reviews	Difference in share of five-star reviews
X_{ij}	Site rating	-0.0176*** (0.0061)	-0.0175 (0.0113)	-0.0083 (0.0102)
	Hotel age	0.0003** (0.0001)	0.00005 (0.0002)	0.0002 (0.0002)
	All suites	0.0086 (0.0065)	-0.0147 (0.0137)	0.0035 (0.0150)
	Convention center	-0.0177** (0.0082)	0.0532*** (0.0147)	-0.0716*** (0.0170)
	Restaurant	0.0376*** (0.0068)	0.0079 (0.0126)	0.0339*** (0.0128)
	Hotel tier controls?	Yes	Yes	Yes
	Hotel location controls?	Yes	Yes	Yes
$OwnAf_{ij}$	Hotel is independent	0.0881*** (0.0079)	-0.0035 (0.0135)	0.0082 (0.0123)
	Multiunit owner	-0.0135** (0.0052)	0.0109 (0.0102)	-0.0239** (0.0117)
Nei_{ij}	Has a neighbor	-0.0091 (0.0080)	0.0285* (0.0156)	-0.0093 (0.0159)
$NeiOwnAf_{ij}$	Has independent neighbor	0.0002 (0.0066)	0.0203 (0.0133)	0.0027 (0.0130)
	Has multiunit owner neighbor	-0.0144** (0.0062)	-0.0150 (0.0125)	-0.0038 (0.0132)
γ_j	City-level fixed effects?	Yes	Yes	Yes
	Observations	2,874	2,874	2,874
	R^2	0.35	0.05	0.07

Notes: Estimation of equation (1) with the sample restricted to hotels that have at least one review by a one-time contributor (the reviewer has only submitted one review on TripAdvisor). The dependent variable in the first column is the share of reviews by one-time contributors among all TripAdvisor reviews for a given hotel. The dependent

TABLE 6—ROBUSTNESS SPECIFICATIONS: SPECIFICATIONS WITH NO CONTROLS

		Difference in share of one- and two-star reviews	Difference in share of one- and two-star reviews	Difference in share of five-star reviews
X_{ij}	Site rating	—	—	—
	Hotel age	—	—	—
	All suites	—	—	—
	Convention center	—	—	—
	Restaurant	—	—	—
	Hotel tier controls	No	No	No
	Hotel location controls	No	No	No
$OwnAf_{ij}$	Hotel is independent	—	0.0093 (0.0092)	0.0429*** (0.0103)
	Multiunit owner	—	-0.0181** (0.0075)	-0.0642*** (0.0084)
Nei_{ij}	Has a neighbor	0.0118 (0.0079)	0.0324*** (0.0098)	-0.0177 (0.0109)
$NeiOwnAf_{ij}$	Has independent neighbor	—	0.0022 (0.0082)	-0.0069 (0.0091)
	Has multiunit owner neighbor	—	-0.0310*** (0.0083)	-0.0211** (0.0093)
γ_j	City level fixed effects?	No	No	No
	Observations	2,931	2,931	2,931
	R^2	0.001	0.01	0.04

Notes: Estimation of equation (1) excluding control variables. The dependent variable in all specifications is the share of reviews that are N star for a given hotel at TripAdvisor minus the share of reviews that are N star at Expedia. Heteroskedasticity robust standard errors in parentheses. All neighbor effects calculated for 0.5 km radius. Results in this table can be compared to the “base” specifications in Table 3 to measure whether and if the inclusion of control variables in Table 3 leads to substantial attenuation of the coefficients of interest.

(a) Table 5: One-Time Contributor Reviewers

(b) Table 6: Specifications with No Controls

7.4 Tables 7 and 8: Negative-review robustness and chain fixed effects

Read:

- Table 7 uses negative reviews as the dependent variable and varies the neighbor radius.
- The multiunit-owner neighbor coefficient is consistently negative and significant across radii.
- Independent neighbor effects are positive in several specifications.
- Table 8 includes chain fixed effects and obtains results similar to baseline estimates.
- Chain fixed effects preserve the key negative-review findings.

Detailed interpretation: Table 7 shows the negative-manipulation result is not driven by an arbitrary 0.5 km radius. Table 8 shows the results survive controls for broad chain-level unobservable preferences and brand effects. Together, these are the key robustness checks for the paper’s strongest result.

Economic message: The evidence for negative manipulation against local competitors is the most robust empirical finding.

TABLE 7—SPECIFICATIONS WITH NEGATIVE REVIEWS AS DEPENDENT VARIABLE

		Difference in share of one-star reviews	Difference in share of one- and two-star reviews	Difference in share of one- and two- star reviews	Difference in share of one- and two-star reviews
X_{ij}	Site rating	-0.0177** (0.0076)	-0.0055 (0.0099)	-0.0050 (0.0099)	-0.0048 (0.0100)
	Hotel age	0.0005*** (0.0001)	0.0003** (0.0002)	0.0003** (0.0002)	0.0003** (0.0002)
	All suites	0.0091 (0.0076)	0.0159* (0.0092)	0.0156* (0.0091)	0.0158* (0.0091)
	Convention center	0.0104 (0.0073)	0.0166* (0.0091)	0.0170* (0.0091)	0.0163* (0.0091)
	Restaurant	0.0039 (0.0076)	0.0111 (0.0092)	0.0091 (0.0092)	0.0091 (0.0092)
	Hotel tier controls?	Yes	Yes	Yes	Yes
	Hotel location controls?	Yes	Yes	Yes	Yes
$OwnAf_{ij}$	Hotel is independent	0.0117 (0.0100)	0.0126 (0.0110)	0.0113 (0.0109)	0.0114 (0.0109)
	Multiunit owner	-0.0025 (0.0047)	-0.0012 (0.0063)	-0.0015 (0.0063)	-0.0020 (0.0063)
Nei_{ij}	Has a neighbor	0.0095 (0.0106)	0.0258** (0.0103)	0.0125 (0.0132)	0.0137 (0.0147)
$NeiOwnAf_{ij}$	Has independent neighbor	0.0191** (0.0081)	0.0109 (0.0099)	0.0176* (0.0095)	0.0207** (0.0093)
	Has multiunit owner neighbor	-0.0204*** (0.0075)	-0.0262*** (0.0084)	-0.0252*** (0.0093)	-0.0262*** (0.0096)
γ_j	City-level fixed effects?	Yes	Yes	Yes	Yes
	Observations	2,931	2,931	2,931	2,931
	Neighbor radius?	0.5 km	0.3 km	0.7 km	0.9 km
	R^2	0.09	0.05	0.05	0.05

(a) Table 7: Negative Reviews Specifications

TABLE 8—CHAIN FIXED EFFECTS ROBUSTNESS CHECK

		Difference in share of one- and two-star reviews	Difference in share of one- and two-star reviews	Difference in share of five-star reviews
X_{ij}	Site rating	-0.0007 (0.0100)	-0.0067 (0.0101)	-0.0193** (0.0089)
	Hotel age	0.0003 (0.0002)	0.0002 (0.0002)	-0.00006 (0.0002)
	All suites	0.0107 (0.0090)	0.0112 (0.0091)	0.0097 (0.0123)
	Convention center	0.0185** (0.0090)	0.0193** (0.0092)	-0.0263** (0.0113)
	Restaurant	0.0085 (0.0095)	0.0077 (0.0095)	0.0271*** (0.0100)
	Hotel tier controls?	Yes	Yes	Yes
	Hotel location controls?	Yes	Yes	Yes
	Chain-level fixed effects?	Yes	Yes	Yes
$OwnAf_{ij}$	Hotel is independent	—	0.0053 (0.0135)	0.0079 (0.0119)
	Multiunit owner	—	0.0053 (0.0067)	-0.0194** (0.0086)
Nei_{ij}	Has a neighbor	0.0205** (0.0096)	0.0304*** (0.0118)	-0.0121 (0.0119)
$NeiOwnAf_{ij}$	Has independent neighbor	—	0.0162* (0.0094)	-0.0071 (0.0099)
	Has multiunit owner neighbor	—	-0.0253*** (0.0088)	-0.0018 (0.0097)
γ_j	City-level fixed effects?	Yes	Yes	Yes
	Observations	2,931	2,931	2,931

(b) Table 8: Chain Fixed Effects Robustness Check

8 Theoretical result blocks

8.1 Result block 1: Cost of manipulation

The appendix model implies that optimal fake-review effort is decreasing in reputational or manipulation cost. In the notation of the paper’s simple model,

$$e_{A,A}^* = \frac{p_A}{2\delta_A t}, \quad e_{A,B}^* = \frac{p_A}{2\gamma_A t}.$$

Thus, when δ_A or γ_A is lower, firm A posts more fake positive reviews for itself or fake negative reviews against B.

Interpretation: This result maps directly into organizational form: independent/single-unit hotels are predicted to manipulate more because reputational costs are lower.

8.2 Result block 2: Negative manipulation requires nearby substitutes

The paper predicts that negative fake reviews are more profitable when competitors are close substitutes. Hotels within a small radius are plausible substitutes, so neighbor variables identify incentives to attack rivals.

Economic message: Competition creates a motive for fake negative reviews. The empirical analysis therefore focuses on nearby competitors and their organizational form.

8.3 Result block 3: Platform verification

Verification raises the cost of fakery. Therefore, the same hotel should show more manipulation-sensitive review patterns on TripAdvisor than on Expedia.

Economic message: Platform design is a policy tool. Verification reduces manipulation but may reduce review volume.

9 Result map

- **Platform gap:** TripAdvisor is more open and has more reviews; Expedia is purchase verified and has fewer reviews.
- **Positive manipulation:** independent and small-owner hotels have more five-star TripAdvisor reviews relative to Expedia.
- **Negative manipulation:** hotels with nearby independent/small-owner competitors receive more negative TripAdvisor reviews relative to Expedia.
- **Organizational discipline:** multiunit owners and large management companies are associated with less manipulation.
- **Suspicious users:** one-time contributors are more common where manipulation incentives are high.
- **Robustness:** no-controls, radius, chain fixed effects, and Orbitz comparisons support the main interpretation.

10 Short summary

- The paper studies promotional online reviews in hotel markets.
- It exploits the difference between TripAdvisor’s open posting and Expedia’s verified-booking review policy.
- The empirical strategy compares TripAdvisor-minus-Expedia review shares across hotel types and local competition conditions.
- Independent hotels and single-unit owners have stronger incentives to fake positive reviews.
- Hotels with nearby independent competitors are more likely to receive negative TripAdvisor reviews.
- Multiunit owners and large management companies appear to curb manipulation.
- The strongest evidence is for negative manipulation against nearby competitors.
- Verification rules reduce manipulation but also reduce review volume.

11 Conclusion

11.1 Core conclusion

Mayzlin, Dover, and Chevalier provide evidence that promotional reviewing is economically meaningful and varies with incentives. Hotels with lower reputational costs and stronger competitive incentives show review distributions consistent with manipulation on the less-verified platform.

11.2 Economic intuition

A fake review is a low-cost advertising message disguised as consumer information. The gain from manipulation is higher when the hotel can benefit from shifting consumers toward itself or away from nearby substitutes. The cost is higher when the organization has a valuable brand, repeated-market reputation, or multiunit spillover risk.

11.3 Incremental contribution revisited

The paper’s incremental contribution is methodological and substantive. Methodologically, it detects hidden manipulation without identifying individual fake reviews. Substantively, it shows that organizational form and local competition shape digital misconduct. This connects online reputation systems to industrial organization and firm-boundary economics.

11.4 Implications for review platforms

Platforms should consider both authentication and review quantity. Verified-purchase requirements reduce fake reviews, but they can dramatically reduce review volume. Hybrid systems that label verified reviews may help, but they do not fully solve aggregation and weighting problems.

11.5 Limitations

The paper infers manipulation rather than observing it directly. The empirical strategy assumes that TripAdvisor and Expedia users do not differentially value hotel ownership and neighbor characteristics in ways that mimic the results. The paper also cannot measure the effect of fake reviews on actual bookings or consumer welfare.

11.6 Takeaway

The central takeaway is that online review manipulation is an incentive problem. When platforms make fakery cheap and firms have strong economic incentives, review distributions shift predictably. Market design and organizational form both matter for review integrity.